

Where the Error Bars on the Red CMX Curve Come From

Short answer

In the LiH absolute-energy-error plot, the **red circles and dashed curve** are energies obtained after connected moments expansion (CMX) error mitigation. Their vertical error bars represent uncertainty caused by finite statistical sampling of the quantum measurements. They are not the difference between the red point and the exact ground-state energy.

Quantities measured for CMX

CMX estimates the ground-state energy from three Hamiltonian moments:

$$a = \langle \hat{H} \rangle, \quad b = \langle \hat{H}^2 \rangle, \quad c = \langle \hat{H}^3 \rangle. \quad (1)$$

Each expectation value is estimated from a finite number of measurement shots. Consequently, a , b , and c have statistical variances $\text{Var}(a)$, $\text{Var}(b)$, and $\text{Var}(c)$.

Propagation into the CMX energy

The CMX energy is a nonlinear function

$$E_{\text{CMX}} = f(a, b, c). \quad (2)$$

The paper propagates the variances of the measured moments through this function:

$$\text{Var}(E_{\text{CMX}}) = \left(\frac{\partial E_{\text{CMX}}}{\partial a} \right)^2 \text{Var}(a) + \left(\frac{\partial E_{\text{CMX}}}{\partial b} \right)^2 \text{Var}(b) + \left(\frac{\partial E_{\text{CMX}}}{\partial c} \right)^2 \text{Var}(c). \quad (3)$$

The plotted error-bar magnitude is therefore the propagated standard uncertainty

$$\boxed{\sigma_{E_{\text{CMX}}} = \sqrt{\text{Var}(E_{\text{CMX}})}}. \quad (4)$$

Thus, a red point at energy E_{CMX} is displayed with a vertical range conventionally interpreted as $E_{\text{CMX}} \pm \sigma_{E_{\text{CMX}}}$.

Equation (3) is Eq. (72) in the paper's Supplementary Information. Its expression specialized to the second-order CMX estimator is given in Supplementary Eq. (73). The authors state that the CMX error bar is calculated using Eq. (73). Because this propagation omits covariance terms, they also note that the actual variance could be smaller.

Values for the LiH red curve

The paper’s *Source Data Fig. 3* spreadsheet labels these numbers “REM+CF+CMX error bar.” For the LiH panel, their magnitudes are:

Bond distance (Å)	Error-bar magnitude (mHa)
0.5	1.747
0.7	4.135
0.9	1.496
1.2	5.452
1.5	1.525
1.8	1.711
2.2	1.615
2.6	2.280
3.0	1.843

Here $1 \text{ mHa} = 10^{-3} \text{ Hartree}$. These values quantify statistical precision. By contrast, the vertical coordinate in the absolute-error panel is

$$|E_{\text{CMX}} - E_{\text{exact}}|, \tag{5}$$

which quantifies agreement with exact diagonalization. These are two different notions of error.

Source

S. Guo *et al.*, “Experimental quantum computational chemistry with optimised unitary coupled cluster ansatz,” *Nature Physics* **20**, 1240–1246 (2024), doi:10.1038/s41567-024-02530-z. See Supplementary Eqs. (72)–(73) and Source Data Fig. 3.